

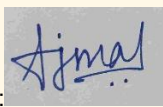
SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DATA CLEANING CHALLENGE

Course Code:	DSA05	Course Title:	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO3	Assessment:	Tool 2 – Data Cleaning Challenge
Weightage:		Total Marks:	25 Marks
CO3 Statement: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex. (BL5 and BL6).			
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Section / Batch:	2023-2027	Date of Submission:	14-08-2026
Faculty In-charge:	K. Veena Devi	Project Mode:	Individual Submission

Code of Conduct

I AJMAL RAHMAN A (192324094) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: 

Anomaly Detection (5 Marks)	Cleaning Methods (5 Marks)	Data Transformation (5 Marks)	Analytical Comparison (5 Marks)	Governance Discussion (5 Marks)	Total (25 Marks)

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks

1. Identify and classify all data quality issues. (5 Marks)

Issue

Missing Customer ID for Sneha

C105 appears twice

Age = 150

Age = 0

Type

Missing value

Duplicate

Outlier/Invalid value

Invalid value

Issue	Type
Missing/incorrect age	Missing value
Quantity = -2	Invalid value
INR, USD and EUR	Inconsistent format/unit
Chennai, chennai, CHENNAI	Inconsistent categorical value
2025-13-01	Invalid date
Tiruchirappalli vs other city formats	Formatting inconsistency

2. Design a cleaning strategy for each issue. (5 Marks)

- Replace missing Customer IDs only if the original transaction can be verified; otherwise mark it as unknown.
- Remove exact duplicate C105 transaction.
- Treat age 0 and 150 as invalid; replace using verified customer information or impute using an appropriate age statistic.
- Negative quantity should be corrected using the original transaction record; do not simply convert it to positive without verification.
- Convert all currencies into a common currency such as INR using a documented exchange-rate table.
- Convert city names to a standard format such as **Chennai, Coimbatore, Madurai, Trichy, Salem.**
- Validate purchase dates using a standard DD-MM-YYYY or ISO format.

3. Perform data standardization and transformation. (5 Marks)

chennai → Chennai
 CHENNAI → Chennai
 INR → INR
 USD → Convert to INR
 EUR → Convert to INR
 10-01-2025 → 2025-01-10
 Revenue=Quantity×UnitPrice

4. Analyze how cleaning affects total revenue calculations. (5 Marks)

Before cleaning, incorrect values can produce misleading revenue.

For example, C104 has:

Therefore:

This produces an incorrect negative revenue.

The duplicate C105 transaction also causes revenue to be counted twice.

5. Generate a report discussing business risks if the data were not cleaned. (5 Marks)

If the data is not cleaned:

- Incorrect customer segmentation
- Wrong product performance ranking
- Incorrect revenue calculations

- Incorrect sales forecasts
- Misleading management reports
- Poor inventory decisions
- Financial losses
- Incorrect business strategies

Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101
NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabtes	15000	D103
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepak N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

Data Issues

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks

1. Detect all inconsistencies and anomalies. (5 Marks)

Issue	Example	Classification
Missing Patient ID	Kavya R	Missing

Issue	Example	Classification
Duplicate record	P1005	Duplicate
Discharge before admission	P1002	Date anomaly
Missing treatment cost	P1005	Missing
Diabetes, diabetes	Diagnosis	Formatting
Diabtes	Diagnosis	Typographical error
HYPERTENSION	Diagnosis	Case inconsistency
Age = 125	Arun V	Outlier
Different date formats	Dates	Formatting

2. Develop rules for handling missing and invalid values. (5 Marks)

- Missing Patient ID → verify from hospital records.
- Missing treatment cost → retrieve from billing system.
- Age > 120 → verify and correct; otherwise mark as invalid.
- Discharge date before admission → verify hospital record.
- Missing dates → retrieve from admission system.
- Duplicate records → retain one valid record and remove the duplicate.

3. Create a process to standardize diagnosis names. (5 Marks)

diabetes → Diabetes
 Diabtes → Diabetes
 HYPERTENSION → Hypertension
 Hypertension → Hypertension
 Asthma → Asthma

4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (5 Marks)

The invalid P1002 record has a discharge date before admission, producing a negative stay duration.

$$\textit{StayDuration} = \textit{DischargeDate} - \textit{AdmissionDate}$$

After correction/removal, the average stay becomes meaningful.

Similarly, the missing treatment cost for P1005 should be recovered or excluded from the average until corrected.

5. Recommend data governance measures to prevent future errors. (5 Marks)

- Use mandatory Patient ID fields.
- Apply date validation rules.
- Maintain a diagnosis master table.
 - Use controlled vocabulary for medical diagnoses.
- Add duplicate detection.
 - Restrict age to realistic ranges.

- Maintain audit logs.

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues

- Missing credit scores
- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks

1. Identify critical issues that could impact loan decisions. (5 Marks)

Major issues:

- Missing credit score for L003
- Missing income for L010
- 900K vs numerical income values
- Negative loan amount for L004
- Duplicate L008
- Age 16 for L005
- FT, Full Time, fulltime
- Extremely high income of ₹75,000,000 for L007

These can directly affect credit-risk calculations and approval decisions.

2. Propose cleaning techniques for missing and inconsistent data. (5 Marks)

Issue	Technique
Missing credit score	Retrieve from credit bureau
Missing income	Verify financial documents
900K	Convert to 900000
Negative loan amount	Verify and correct
Duplicate L008	Remove duplicate
Age < 18	Flag as invalid
FT/fulltime/Full Time	Standardize as Full Time
Extreme income	Investigate and validate

3. Detect and justify treatment of outliers. (5 Marks)

The income of: ₹75,000,000

is extremely high compared with most other records.

4. Analyze the effect of cleaning on loan approval rates. (5 Marks)

Cleaning can change approval rates because invalid records may incorrectly influence decisions.

For example:

- Duplicate approved records can artificially increase approvals.
- Invalid age may create an inappropriate approval.
- Missing credit score may require the application to be reviewed instead of automatically approved.
- Incorrect income can distort eligibility.

5. Discuss ethical implications of incorrect data cleaning decisions. (5 Marks)

Incorrect cleaning can create unfair decisions.

For example:

- Removing genuine high-income customers can create bias.
- Incorrectly imputing credit scores can change someone's eligibility.
- Automatically rejecting records with missing values can disadvantage certain customers.
- Gender, age, or socioeconomic characteristics should not be unfairly used.

Therefore, cleaning must be transparent, explainable, consistent, and auditable.

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

Tasks

1. Assess the quality of the traffic dataset. (5 Marks)
 1. Assess the dataset quality

Important issues include:

- S105 has a missing timestamp.
- S106 is duplicated.
- S103 has vehicle count = -50.
- S104 has speed = 350 km/h.
- S106 has missing weather information.
- Anna Salai, anna salai, ANNA SALAI are inconsistent.
- Old Mahabalipuram Road is split across lines/formats.
- S10A has an ID formatting error.

Therefore, the dataset requires substantial cleaning before traffic analysis.

2. Recommend cleaning techniques for time-series data. (5 Marks)

Sort records chronologically.

Standardize timestamps.

Detect missing timestamps.

Remove duplicate readings.

Interpolate suitable missing numerical values when justified.

Use forward/backward filling carefully for categorical values.

Detect sudden unrealistic changes.

Maintain consistent sampling intervals.

3. Design rules to identify faulty sensor readings. (5 Marks)

Vehicle count < 0 → Invalid

Average speed < 0 → Invalid

Average speed > 300 km/h → Flag for investigation

Missing timestamp → Flag

Duplicate Sensor ID + Timestamp → Duplicate

Invalid Sensor ID format → Flag

4. Evaluate how cleaning affects congestion analysis. (5 Marks)

A congestion analysis can use:

$$\text{Congestion} \propto \frac{\text{VehicleCount}}{\text{AverageSpeed}}$$

Invalid readings can produce incorrect congestion levels.

For example, a speed of **350 km/h** would make the road appear unusually uncongested even though the sensor reading is unrealistic.

After cleaning:

- False congestion decreases.
- Traffic trends become reliable.
- Peak traffic periods can be identified accurately.
- Traffic signal optimization becomes more effective.

5. Suggest automated monitoring techniques for future data collection. (5 Marks)

Implement:

- Real-time validation
- Range checks
- Duplicate detection
- Sensor health monitoring
- Missing-data alerts
- Timestamp validation
- Automated anomaly detection
- Data-quality dashboards
- Automatic logging of sensor failures

Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5
UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (5 Marks)

Step 1 – Data profiling

Identify:

- Missing values
- Duplicates
- Outliers
- Invalid values
- Formatting differences

Step 2 – Data validation

Check:

- SalesAmount > 0
- CustomerRating between 1 and 5
- Valid dates
- Valid currencies
- Valid country/region combinations

Step 3 – Standardization

Standardize:

- Currency
- Product names
- Salesperson names
- Dates
- Region and country names

Step 4 – Missing-value handling

Use:

- Source verification
- Business rules
- Statistical imputation where appropriate

Step 5 – Outlier detection

Use:

- IQR
- Z-score
- Domain knowledge

Step 6 – Final validation

Recheck the cleaned dataset before analysis.

2. Design methods to standardize currencies, products, and dates. (5 Marks)

INR → Common currency

USD → Common currency

EUR → Common currency

GBP → Common currency

3. Handle missing values and outliers using appropriate analytical techniques. (5 Marks)

Missing sales amount

Priya S's Mobile transaction has a missing SalesAmount.

Possible approaches:

1. Retrieve the original sales record.
 2. If unavailable, use a documented imputation method.
 3. If reliable imputation is impossible, exclude it from revenue calculations and flag it.
4. Compare regional sales rankings before and after cleaning. (5 Marks)
Before cleaning, the ₹9,999,999 transaction can artificially inflate Asia's sales ranking.
The duplicate European transaction also artificially increases Europe's sales.

Therefore:

Before cleaning

Regional rankings may be distorted by:

- Duplicate sales
- Extreme outliers
- Multiple currencies
- Missing values

After cleaning

Regional sales should be recalculated using:

Then rank:

1 → Highest regional sales

2 → Second highest

3 → Third highest

...

The final ranking should be based only on **valid, standardized, non-duplicate sales records**.

5. Prepare an executive report explaining how data quality influences strategic business decisions. (5 Marks)

Poor-quality sales data can lead to incorrect strategic decisions.

Major business risks:

- Wrong regional performance ranking
- Incorrect sales forecasts
- Incorrect product decisions
- Poor sales-person performance evaluation

- Incorrect revenue reporting
- Wrong market expansion decisions
- Financial losses

For example, an extreme sales value of ₹9,999,999 can make a region appear much more successful than it actually is. Similarly, duplicate transactions can artificially increase regional revenue.